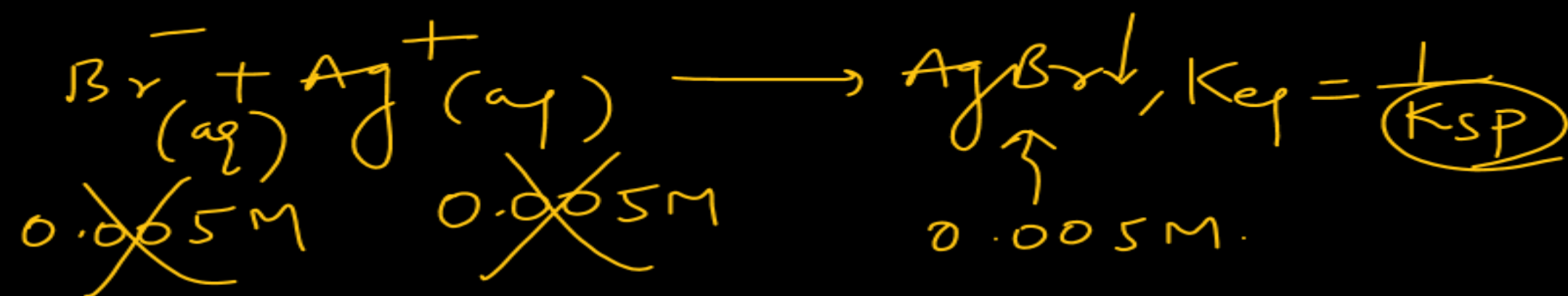


Good

250ml, 0.02M NaCl(aq) + 250 ml, 0.02M NaBr(aq) + 500 ml, 0.01M AgNO₃.

Find equilibrium concentration of Br⁻ in mol/lit.

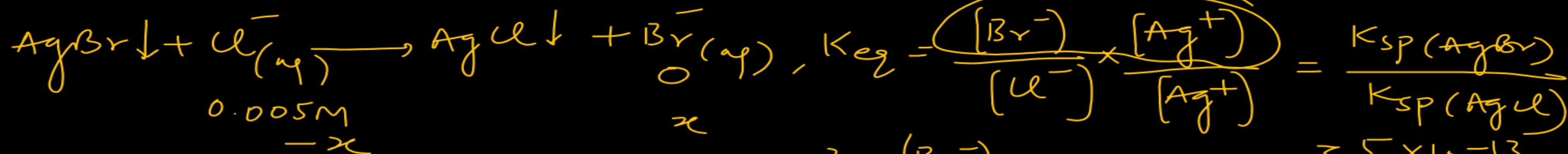
$K_{sp}(\text{AgCl}) = 10^{-10}$, $K_{sp}(\text{AgBr}) = 5 \times 10^{-13}$



$$[\text{Cl}^-]_0 = \frac{0.02 \times 0.25 \text{ mol}}{1 \text{ lit}} = 0.005\text{M}$$

$$[\text{Br}^-]_0 = \frac{0.02 \times 0.25}{1} = 0.005\text{M}$$

$$[\text{Ag}^+]_0 = \frac{0.01 \times \frac{500}{1000}}{1} = 0.005\text{M}$$



$$\begin{aligned} K_{eq} &= 5 \times 10^{-3} = \frac{[\text{Br}^-]}{[\text{Cl}^-]} = \frac{x}{0.005 - x} \\ 25 \times 10^{-6} - 5 \times 10^{-3} x &= x \\ (1 + 5 \times 10^5) x &= 2.5 \times 10^{-5} \Rightarrow \boxed{x = 2.5 \times 10^{-5} \text{ Molar}} \end{aligned}$$
$$\begin{aligned} &= \frac{5 \times 10^{-13}}{10^{-10}} \\ &= 5 \times 10^{-3} \end{aligned}$$

The critical temperature and pressure of a Vander Waals gas is -177°C and 20 atm, respectively. At Boyle's temperature, the gas behaves ideally up to 50 atm. If 0.2 moles of this gas is taken at the temperature and pressure given in Column I, then match with the expected volume of system in Column II ($R = 0.08 \text{ L-atm/K-mol}$).

Column-I

(A) -177°C , 20 atm

(B) 51°C , 6.48 atm

(C) 77°C , 7.0 atm

(D) 27°C , 6.0 atm

(E) 51°C , 64.8 atm

(B) $Z=1$, $V_{\text{real}} = V_{\text{ideal}}$

A

A	→	Q
B	→	S
C	→	P
D	→	R
E	→	T

Column II

(P) 821 ml

(Q) 28.8 ml

(R) 760 ml

(S) 800 ml

(T) 85 ml

$$T_c = -177^{\circ}\text{C}$$

$$= (273 - 177) \text{K} = 96 \text{K}$$

$$P_c = 20 \text{ atm}$$

$$T_b = \frac{a}{Rb} \quad \bigg| \quad T_c = \frac{8a}{27Rb} = \frac{8}{27} \times T_b$$

$$= \frac{27}{8} \times T_c = \frac{27}{8} \times 96 = 324 \text{K}$$

$$V_c = 3b$$

$$\frac{T_c}{P_c} = \frac{8a}{27Rb}$$

$$= \frac{8b}{R} = \frac{96}{20}$$

$$b = \frac{96}{20} \times \frac{R}{8} = \frac{3}{5} \times R$$

(A)

$$V_c = 3 \times \frac{3}{5} \times 0.08 \text{ lit} = 3 \times \frac{5}{5} \times 0.08 \times 1000 \text{ ml} = 9 \times 16 \text{ ml}$$

$$PV = nRT \Rightarrow V_{\text{id}} = \frac{0.2 \times 0.08 \times 324}{6.48} \times 1000 \text{ ml} = 800 \text{ ml}$$

$$V = 0.2 \times 9 \times 16 = 3.2 \times 9 = 28.8$$

(Q)

(S)

(P)

(R)

(T)

(C) $77^{\circ}\text{C} > T_b$

$z > 1, V_{\text{real}} > V_{\text{ideal}}$

$$V_{\text{id}} = \frac{nRT}{P} = \frac{0.2 \times 0.08 \times 350 \times 1000}{1} \text{ ml}$$

$$= 2 \times 8 \times 50 = 800 \text{ ml}$$

$V_{\text{real}} > 800 \text{ ml}$

(D) $27^{\circ}\text{C}, 6 \text{ atm}, V_{\text{id}} = \frac{nRT}{P} = \frac{0.2 \times 0.08 \times 300 \times 1000}{6}$

$$= 800 \text{ ml}$$

$V_{\text{real}} < 800 \text{ ml}$

$$\begin{array}{r} 273 \\ 77 \\ \hline 350 \end{array}$$

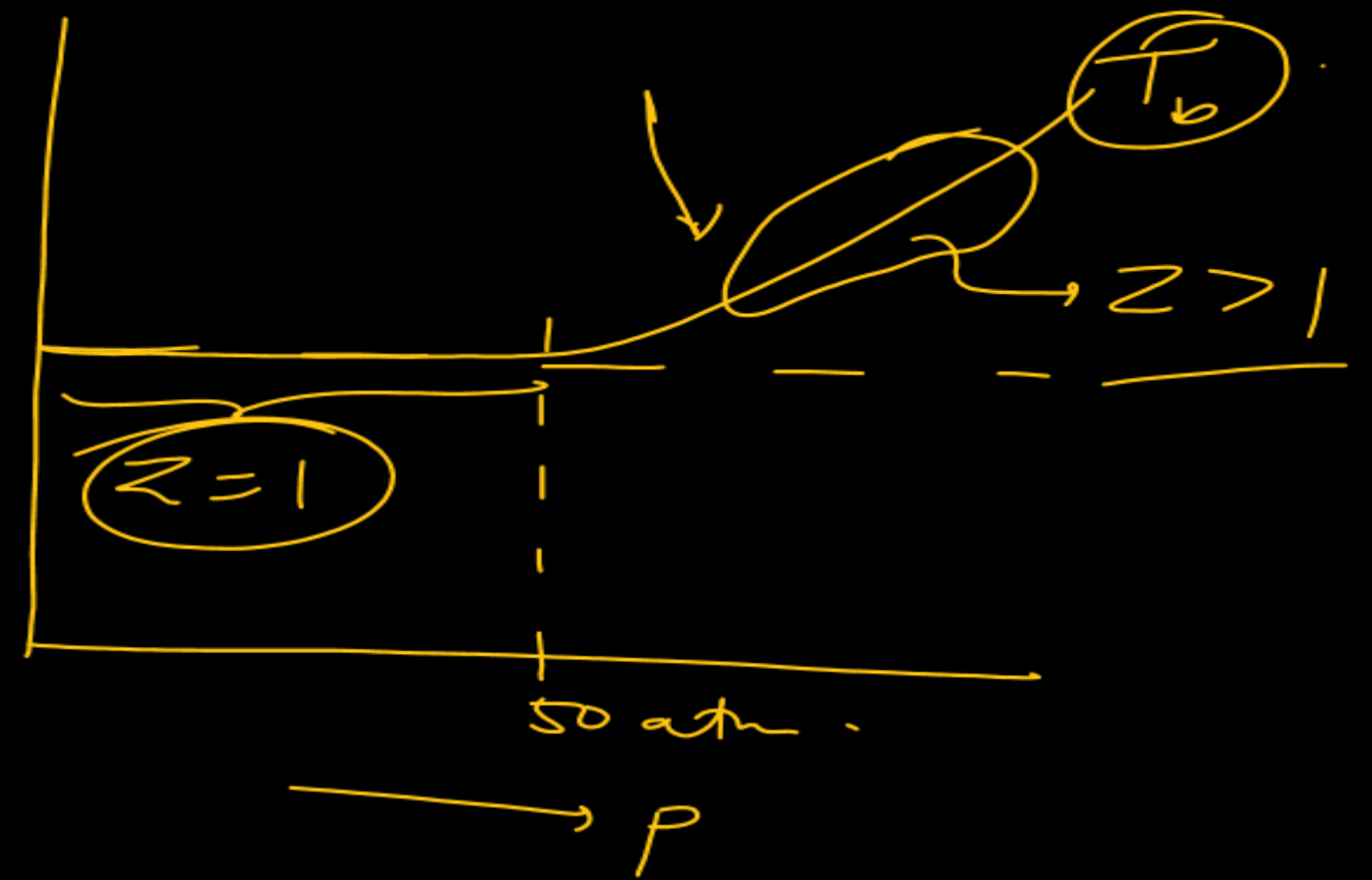
(E) 512, 64.8 atm
 $z > 1$

$$V_{id} = \frac{nRT}{P}$$

$$= \frac{0.2 \times 0.08 \times 324 \times 1000}{64.8}$$

$$= 80 \text{ ml}$$

$$V_{real} > 80 \text{ ml}$$



solubility

The solubility of $H_2(g)$ and $CO(g)$ in molten $KNO_3/NaNO_3$ is given by

$$\log_{10}(S_{H_2}) = -5.4 - 768(T/K)^{-1} \quad \text{or} \quad \log S_{H_2} = -5.4 - \frac{768}{T}$$

$$\log_{10}(S_{CO}) = -6 - 980(T/K)^{-1} \quad \log S_{CO} = -6 - \frac{980}{T}$$

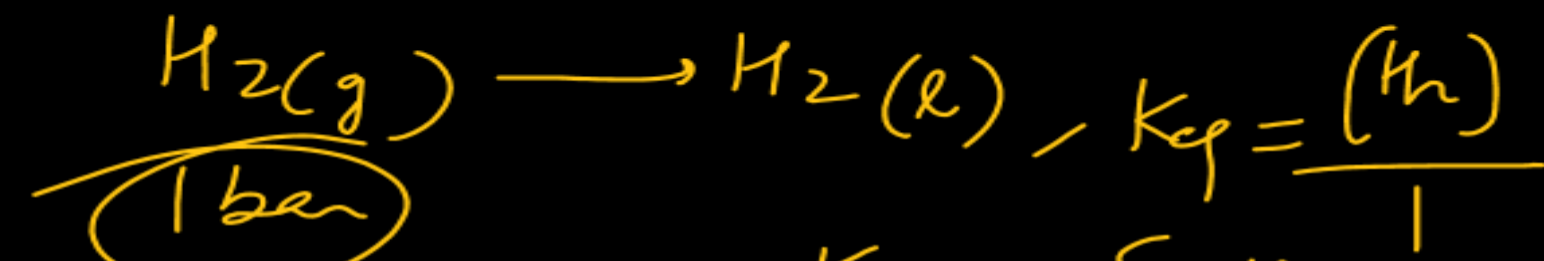
Where S is the solubility in mole cm^{-3} bar $^{-1}$. Based on above write up-identify the correct statement(s) among the following. {Assume ΔS° and ΔH° do not vary with 'T'}

✓ (A) $\Delta H^\circ_{\text{solution}} [CO(g)] > \Delta H^\circ_{\text{solution}} [H_2(g)]$ (at 500K)

✗ (B) $\Delta S^\circ_{\text{solution}} [CO(g)] > \Delta S^\circ_{\text{solution}} [H_2(g)]$ (at 500K)

✗ (C) At very high temperature the process $CO(g) + \text{solvent} \rightarrow CO(\text{solution})$ becomes spontaneous process in standard conditions

(D) A 500 K the molar enthalpy of solution of $H_2(g)$ is nearly 14.7 kJ/mole.



$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -2.3RT \log K_{eq}$$

$$-2.303RT \log S_{H_2} = \Delta H^\circ - T\Delta S^\circ$$

$$\log S_{H_2} = -\frac{\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R}$$

$$\frac{(\Delta H^\circ)_{H_2}}{2.303R} = 768$$

$$(\Delta H^\circ)_{H_2} = 2.3 \times 8.3 \times 768 \text{ J}$$

$$(\Delta H^\circ)_{CO} = 2.3 \times 8.3 \times 980 \text{ J/mole}$$

$$\frac{(\Delta S^\circ)_{H_2}}{2.3 \times 8.3} = -5.4$$

$$(\Delta S^\circ)_{H_2} = -5.4 \times 2.3 \times 8.3 \text{ J/K/mole}$$

$$(\Delta S^\circ)_{CO} = -6 \times 2.3 \times 8.3 \text{ J/K/mole}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$\uparrow$ +ve $\uparrow$ -ve
 $(\Delta H^\circ)_{H_2} = \frac{2.3 \times 8.3 \times 768}{1000} \text{ kJ} = 14.7 \text{ kJ/mole}$

Q.

Match the type of processes involved in the extraction of metal given in column-I with the given ores in column-II

Column-I

- (A) Slag formation
(B) Froth – floatation
(C) Leaching
(D) Roasting

in smelting process
(flux)

p, r, s, t

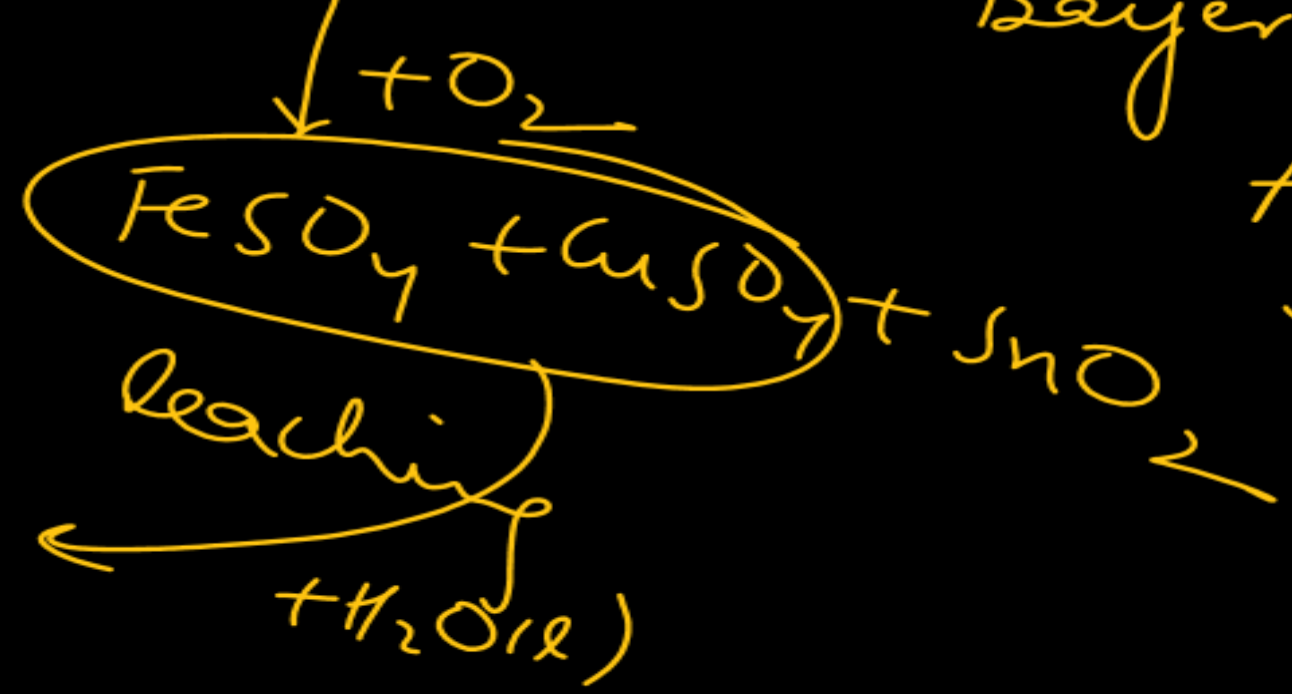
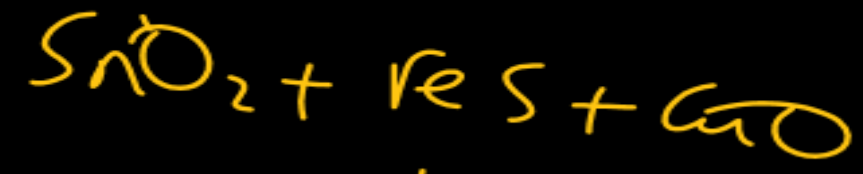
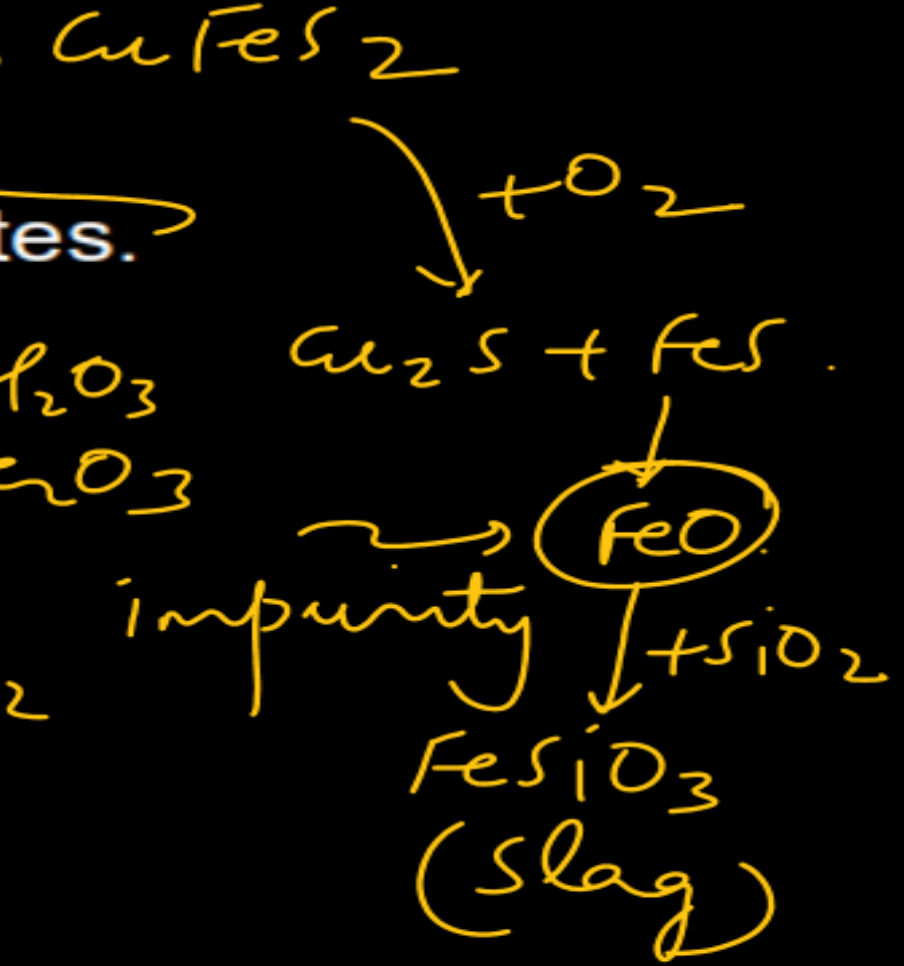
p, t

q, s

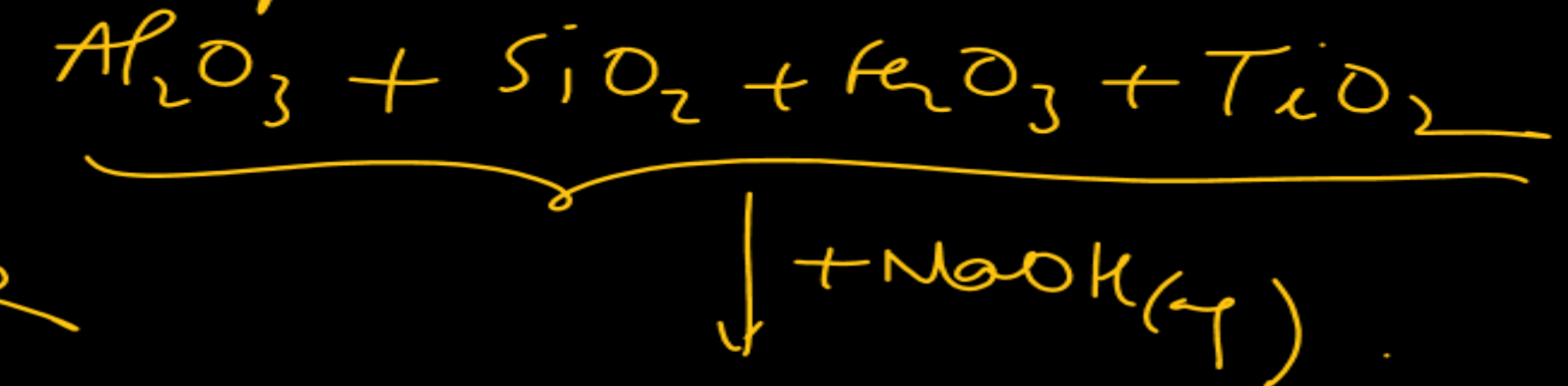
p, r, s, t

Column-II

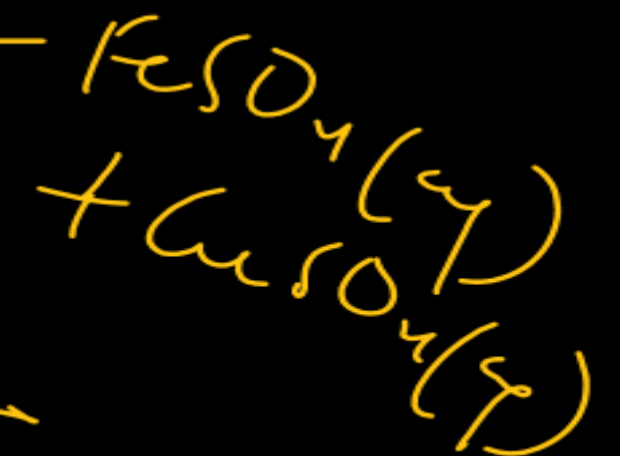
- (p) Extraction of copper from copper pyrites.
(q) Extraction of aluminium from bauxite.
(r) Extraction of iron from haematite.
(s) Extraction of tin from cassiterite
(t) Extraction of lead from galena.



Bayer's process



(Black tin)



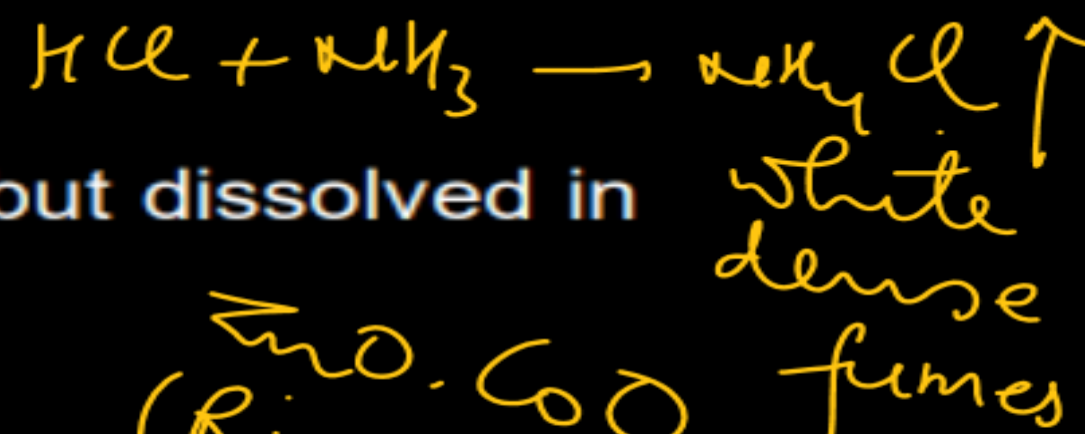
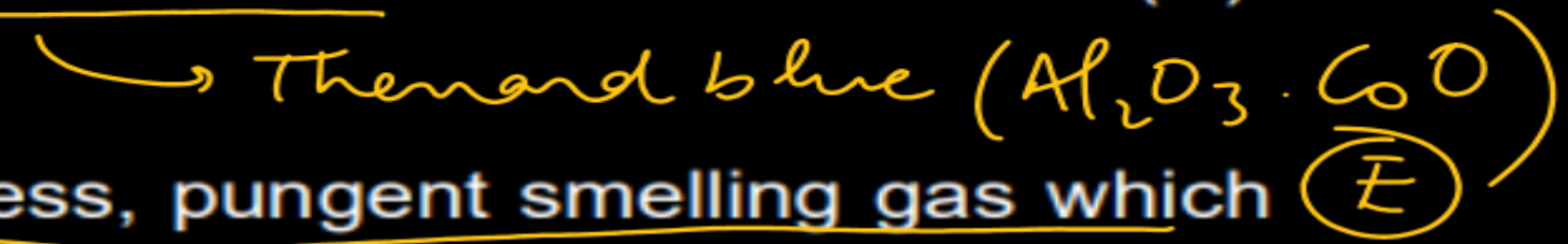
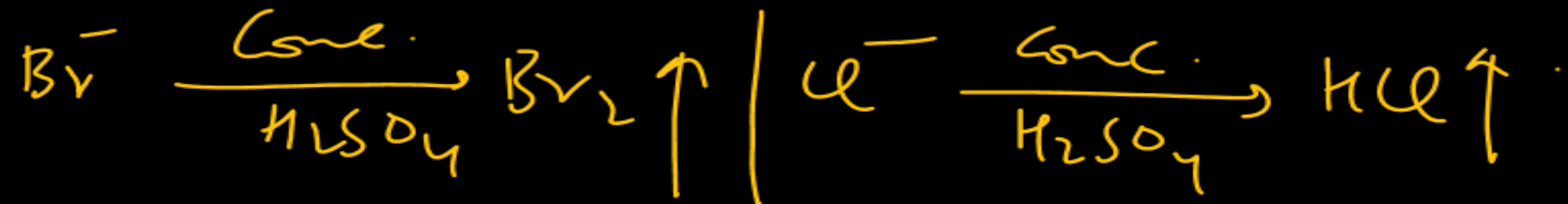
~~70-80%~~
 $\approx 70\%$ of SnO_2

~~Zn^{2+}~~ / Al^{3+} or $AlCl_3$ $\rightarrow BaSO_4$ / ~~$PbSO_4$~~
 A white salt mixture contain two salts, salt A is water soluble and salt B is water insoluble. the salt mixture was leached with water so that soluble portion went into solution, leaving behind residue of B. Solution was filtered off and filtrate was preserved for further analysis. Precipitate B was insoluble in dilute HCl as well as in concentrated ammonium acetate solution but dissolved in boiling solution of concentrated sulphuric acid. The solution was cooled and treated with K_2CrO_4 giving a yellow precipitate (C), that was insoluble in dilute acetic acid solution. Also the original filtrate gave a white gelatinous precipitate (D) on treatment with aqueous ammonia. D was dissolved in small volume of concentrated nitric acid and few drops of cobalt nitrate was added to the above solution. Now a filter paper was dipped into the solution and burned finally to give a blue fixed ash due to formation of (E).

Sodium extract of the original salt mixture gave the following results:

(a) Boiling with concentrated sulphuric acid solution gave off a colourless, pungent smelling gas which formed a white cloud on the surface of a glass rod moistened with ammonia solution.

(b) Gave a white precipitate with lead acetate solution, that was insoluble in dilute acid but dissolved in concentrated solution of ammonium acetate. Identify A to E.



Ans: (4)
 Dissociation of salt $A_2B_3(aq)$ in water follows first – order kinetics $A_2B_3(aq) \rightarrow 2A^{3+}(aq) + 3B^{2-}(aq)$.
 Starting from $t = 0$, a definite volume of solution is taken at different instant and its osmotic rise is measured. If the osmotic rise was 2 mm at $t = 0$ and 6 mm at $t = 10$ min, then the value of $\frac{x}{y}$ is, where

x = osmotic pressure of solution (in Pascal) at $t = 20$ minutes and $y = t_{3/4}$ of the dissociation reaction (in min) [Given : Density of final solution in each case = 1.0 g/ml, $g = 10 \text{ m/s}^2$]



$t_{1/2} = 10 \text{ min}$
 $(i)_{10 \text{ min}} = 1 + (n-1)\alpha$
 $= 1 + (5-1)\alpha$
 $= 1 + 4\alpha$

$t = 20 \text{ min}, \alpha = 0.75, i = 1 + 4 \times 0.75 = \text{(4)}$

$\pi = iCRT = h d g$
 $t = 0, \quad 1 \times C R T = 2 \times d g \quad \text{--- (1)}$

$t = 10, \quad (1+4\alpha) \times C R T = \frac{6}{3} \times d g \quad \text{--- (2)}$
 $1+4\alpha = 3 \Rightarrow \alpha = 0.5$

$(\pi)_{20 \text{ min}} = 4 \times C R T = 4 \times 2 d g$
 $= 8 \times 10^{-3} \times 1 \times 10^3 \times 10$
 $= 80 \text{ Pa}$

$\Rightarrow \frac{x}{y} = 4$

To a 10 ml 1 M aqueous solution of Br_2 , excess of NaOH is added so that all Br_2 is disproportionated to Br^- and BrO_3^- . The resulting solution is freed from Br^- , by extraction and excess OH^- neutralised by acidifying the solution. The resulting solution is sufficient to react with 1.5g of impure CaC_2O_4 ($M = 128$ g/mol) sample. The % purity of Oxalate sample is

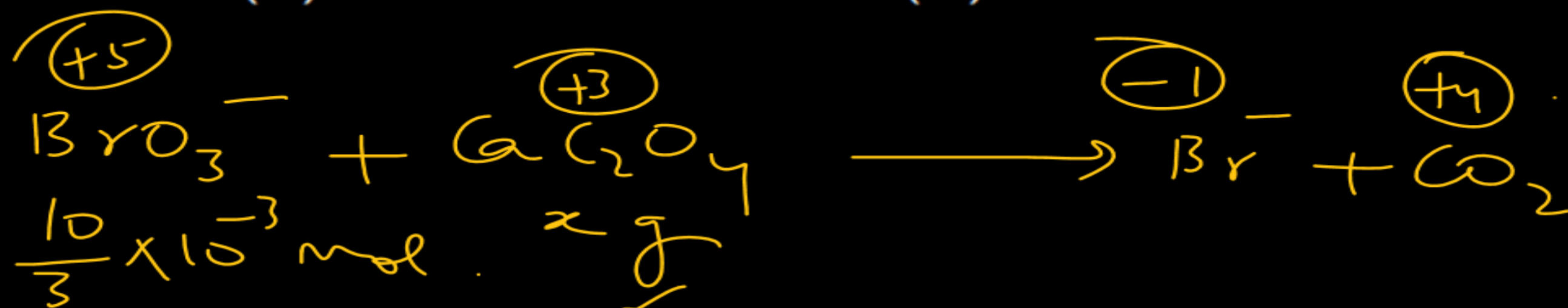
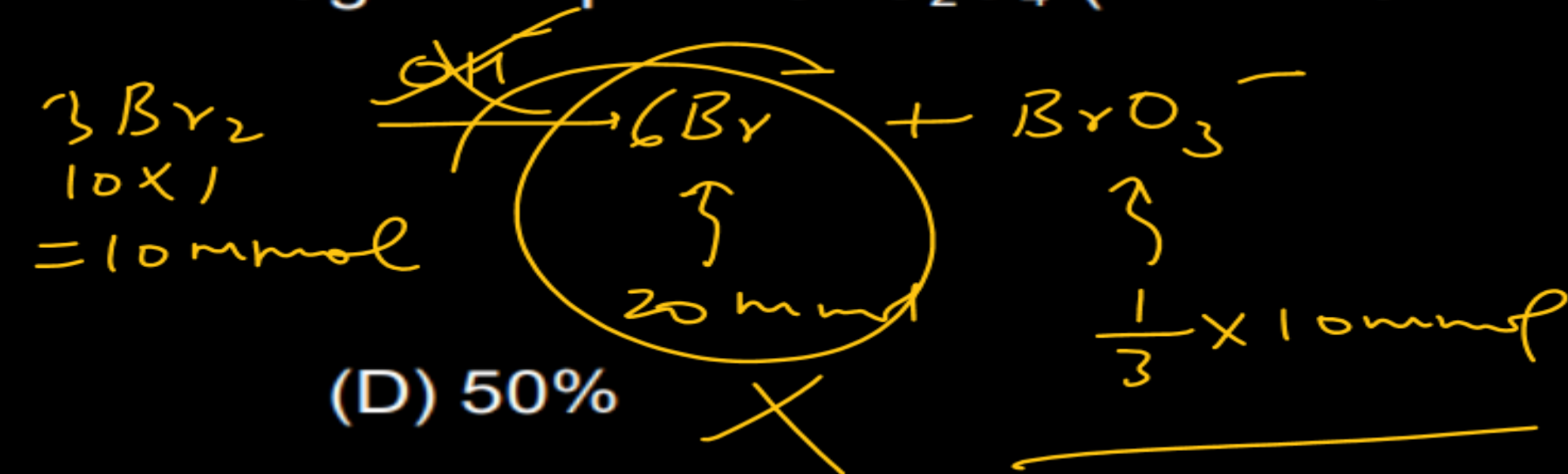
The relevant reactions are $\text{Br}_2(\text{aq.}) + \text{OH}^- \longrightarrow \text{Br}^- (\text{aq.}) + \text{BrO}_3^-$
 $\text{BrO}_3^- + \text{C}_2\text{O}_4^{2-} \longrightarrow \text{Br}^- + \text{CO}_2$

(A) 85.3%

(B) 12.5%

(C) 90%

(D) 50%



$\frac{1}{3} \times 10^{-2} \times 6 = \frac{x}{128} \times 2$

$x = 1.28 \text{ g}$

$\% \text{ Purity of oxalate} = \frac{1.28}{1.5} \times 100 = 85.3\%$

